

# Boğaziçi Üniversitesi



**IE582** Project Report

Instructor: Mustafa G. **Baydoğan**  
Assistant: Abdullah **Kayacan**

Burak Berk **Bulut** - 2024776003  
Oğuzhan **Engin** - 2019402009  
Hasan Yağız **Kılıç** - 2019402147

# Table of Contents

|                                      |           |
|--------------------------------------|-----------|
| <b>Table of Contents.....</b>        | <b>2</b>  |
| <b>Executive Summary.....</b>        | <b>3</b>  |
| <b>Introduction.....</b>             | <b>4</b>  |
| Background.....                      | 4         |
| Objectives.....                      | 4         |
| Scope.....                           | 4         |
| Methods Overview.....                | 4         |
| Challenges.....                      | 5         |
| Significance.....                    | 5         |
| <b>Literature Review.....</b>        | <b>6</b>  |
| <b>Data Overview.....</b>            | <b>7</b>  |
| Dataset Description.....             | 7         |
| Data Preprocessing.....              | 7         |
| <b>Feature Engineering.....</b>      | <b>8</b>  |
| 1. Engineered Features.....          | 8         |
| 2. Directly Selected Features.....   | 9         |
| <b>Methodology.....</b>              | <b>10</b> |
| <b>Results and Analysis.....</b>     | <b>11</b> |
| Classification-Based Prediction..... | 12        |
| Regression-Based Prediction.....     | 15        |
| Feature Importances.....             | 18        |
| Betting Strategy Evaluation.....     | 19        |
| <b>Discussion.....</b>               | <b>21</b> |
| <b>Conclusion.....</b>               | <b>23</b> |
| <b>References.....</b>               | <b>24</b> |

# Executive Summary

This report presents a comprehensive methodology for predicting soccer match outcomes and designing a live betting strategy. Leveraging machine learning algorithms, the study integrates statistics and real-time betting odds to produce actionable insights for decision-making. The project employs models such as Random Forest, and Gradient Boosted Decision Trees to predict match outcomes (home win, draw, away win or goal difference) and determine the most profitable betting opportunities. Since the decision-making process includes a variety of aspects, building an expectation based optimization function for the live betting is the other priority of the study.

The challenges of the project can be separated into 2 related sections. The first key component of the project is the analysis of the data by using statistical learning with using data preprocessing to handle real-time match statistics and odds, feature engineering to derive advanced metrics like shot efficiency and team form, and hyperparameter optimization for model tuning. Last but not least, building an algorithm which helps to decide the betting strategy is the other main target of the project.

Two primary strategies are developed and tested:

1. Ranked Probability Score (RPS)-based parameter tuning for machine learning models predicting match outcomes.
2. Expected values of the returns based betting strategy.

The project employs cross-validation to ensure robust evaluation of models, with performance measured using metrics such as RPS, accuracy, and mean absolute error (MAE). The final results demonstrate the effectiveness of combining machine learning with domain knowledge to optimize live betting strategies, achieving significant predictive accuracy and profitability.

# Introduction

## Background

Predicting the results of soccer matches has long been a challenging task due to the dynamic nature of sports and the numerous factors influencing results. From in-game events like goals and red cards to pre-game metrics such as team form and historical performance, soccer analytics presents a rich and complex domain for machine learning applications. Even though how rich are the inputs, these features are never enough for a robust model. The outcomes are affected by so many aspects which are not parameterized by the inputs. Due to this nature of the sports competitions it has always been a challenge to model these problems.

This project addresses the intersection of sports analytics and live betting by developing a data-driven framework for predicting match outcomes and identifying optimal betting opportunities. By leveraging real-time data and machine learning models, the study aims to replicate market behavior and outperform conventional betting strategies.

## Objectives

The primary objectives of this project are:

1. **Outcome Prediction:** Accurately predict match results (home win, draw, or away win) using advanced machine learning models.
2. **Betting Strategy Optimization:** Identify the most profitable moments to place bets during a match, considering real-time odds and market probabilities.

## Scope

The study uses a rich dataset comprising detailed match statistics and real-time betting odds. Key elements of the analysis include:

- Preprocessing data to account for time order and forward-filling missing values.
- Engineering features like shot efficiency, attack-defense ratios, and team momentum to enhance predictive power.

The models are trained and tested on historical match data, with a focus on ensuring time-consistent splits to prevent forward-looking bias. Simulated betting strategies are evaluated on a test set to assess both accuracy and profitability.

## Methods Overview

1. **Data Preparation:**
  - Sorting and grouping match data by time order to preserve chronological integrity.

- Forward-filling and imputing missing values to create a consistent dataset.
- 2. **Feature Engineering:**
  - Deriving team-specific metrics such as form, streaks, and weighted performance.
  - Calculating advanced statistics like shot efficiency, expected goals, and game pressure for each team.
- 3. **Modeling Approaches:**
  - Classification models (Random Forest, Gradient) for predicting match outcomes.
  - Regression models for predicting goal differences as a basis for betting decisions.
  - Time-series cross-validation is applied to tune both classification and regression models.
- 4. **Evaluation Metrics:**
  - **Ranked Probability Score (RPS):** Measures the quality of probabilistic predictions.
  - **Expected Return:** Assess the total profit from simulated bets and identify profitable patterns.

## Challenges

The project addresses several challenges inherent in sports analytics:

- Handling noisy and incomplete data, especially for real-time match statistics.
- Designing a betting strategy that accounts for dynamic market odds and probabilistic predictions.
- Avoiding forward-looking bias by ensuring predictions are based solely on available information at the time of decision-making.

## Significance

This study contributes to the growing field of sports analytics by demonstrating how machine learning can be applied to optimize live betting strategies. The integration of domain-specific knowledge with robust predictive models not only improves match outcome predictions but also highlights the potential for data-driven decision-making in real-world betting markets.

# Literature Review

Predicting soccer match outcomes and developing betting strategies have long been of significant interest within the fields of sports analytics, machine learning, and predictive modeling. Recent research has explored various methods to improve prediction accuracy, focusing on statistical models, machine learning approaches, and domain-specific feature engineering.

One of the earliest and most influential approaches to predicting soccer outcomes was based on statistical modeling. Maher (1982) introduced a Poisson regression model to predict match results by estimating the number of goals scored by each team. This approach relied on historical match data, assuming that goals follow a Poisson distribution. The model demonstrated reasonable predictive power, especially in terms of predicting the number of goals rather than binary outcomes like win, loss, or draw.

Berrar et al. (2019) highlighted the importance of feature engineering in machine learning models for soccer prediction. Their study introduced two novel feature engineering methods: "recency feature extraction" and "rating feature learning." The recency feature extraction method captures a team's recent form by considering performance metrics over a sliding time window. In contrast, rating feature learning assigns dynamic ratings to teams based on their attacking and defensive capabilities, which are updated after each match.

Horvat and Job (2020) conducted an extensive review of machine learning algorithms applied to sports outcome prediction, analyzing over 100 papers. They observed that neural networks, particularly deep learning models, were among the most frequently used algorithms. However, they noted that traditional models like logistic regression and SVM still performed competitively when combined with effective feature selection and extraction.

Baboota and Kaur (2019) explored the use of random forests and gradient boosting for predicting outcomes in the English Premier League. Their work emphasized the role of feature selection in improving model performance. By selecting key features such as goal differences, team strength ratings, and recent match statistics, their models outperformed several baseline approaches. However, their models, while accurate, were unable to consistently outperform bookmakers' odds due to the inherent unpredictability of soccer matches. The weighted streak and team form features from this study are integrated into our model and discussed in the methodology section.

Horvat et al. (2020) discussed the importance of modeling dynamic odds in live betting. They noted that in-game events such as goals, red cards, and substitutions significantly impact the odds offered by bookmakers. Traditional machine learning models, which rely on static pre-match data, are insufficient for live betting scenarios. Instead, models need to incorporate streaming data and update predictions continuously throughout the match.

# Data Overview

This section provides an overview of the dataset used for the study, highlighting its structure, main attributes, and the preprocessing steps performed to prepare it for machine learning models and betting strategy development.

## Dataset Description

The dataset contains detailed information on soccer matches, including real-time match statistics, betting odds, and metadata. It is designed to support dynamic, time-sensitive analysis for predicting match outcomes and optimizing live betting strategies.

Key features of the dataset include:

- **Scope:** Covers matches before November 1, 2024, for training and those after this date for testing, ensuring chronological order and avoiding forward-looking bias.
- **Columns:**
  - **Metadata:** Includes match identifiers (`fixture_id`), start times (`match_start_datetime`), and specific time points (`minute`, `second`).
  - **Odds Data:** Raw betting odds (1, X, 2) and their normalized probabilities.
  - **Match Statistics:** Offensive metrics (e.g., shots on target), defensive metrics (e.g., tackles), and event-based measures (e.g., goals, corners, red cards).

The dataset contains about 50,000 rows, with dynamically updated features capturing both offensive and defensive aspects of the game.

## Data Preprocessing

To ensure data quality and consistency, the following preprocessing steps were applied:

1. **Sorting and Forward-Filling:**
  - Data is sorted by match time to preserve chronological order.
  - Missing values within matches are forward-filled to maintain continuity, while initial missing values (e.g., before play began) were replaced with zeros.
2. **Odds Normalization:**
  - Raw odds are converted into probabilities and normalized to account for bookmaker adjustments, ensuring these probabilities summed to one.
3. **Filtering:**
  - Suspended or stopped matches are excluded, as the associated odds and statistics are deemed unreliable.
4. **Feature Integration:**
  - Derived features like goal difference, attack and defense scores, and team form metrics are added to enrich the dataset.
  - Team-specific historical data is incorporated to capture performance trends.

These steps resulted in a comprehensive dataset of around 150 features per row, ready for predictive modeling and betting simulations. This structured and reliable dataset was critical to achieving the project's objectives.

## Feature Engineering

Feature engineering is extracting new features from the raw data into meaningful variables. As can be seen, this is a critical step for the project since it directly affects the predictive capabilities of the machine learning models. The logic behind the features extracted and selected directly from the raw dataset relies on the content - knowledge. There are 2 main categories under the feature engineering phase.

### 1. Engineered Features

The following features are meticulously engineered using domain knowledge and mathematical transformations. These features aim to capture the dynamic, tactical, and psychological elements of soccer matches.

All the features below are shown for the home teams. However, all of them can be calculated for away teams too.

- Goal Difference: Serves as both a feature (e.g., assessing a team's dominance) and a target for regression-based models predicting the final score.  
$$\text{Goal Difference} = (\text{Goals} - \text{home}) - (\text{Goals} - \text{away})$$
- Shot Efficiency: Measures how efficiently a team converts total shots into goals. Consider that higher efficiency reflects better finishing quality and is a strong indicator of offensive performance.  
$$\text{Shot Efficiency} - \text{home} = (\text{Goals} - \text{home}) / \text{Shots Total} - \text{home}$$
- Cross Success Rate: The proportion of effective crosses over the total attempts. Indicates the ability to create scoring opportunities through effective wing play.  
$$\text{Cross Success Rate} - \text{home} = (\text{Accurate Crosses} - \text{home}) / \text{Total Crosses} - \text{home}$$
- Shot Distribution (Close Shots): The proportion of the shots taken inside the box. Close range shots have more goal expectations.  
$$\text{Close Shot Distribution} - \text{home} = (\text{Shots Insidebox} - \text{home}) / \text{Shots Total} - \text{home}$$
- Attack to defense Ratio: Compares offensive and defensive effectiveness. Teams with a higher ratio are more likely to dominate the game.  
$$\text{Attack to Defense Ratio} = (\text{Dangerous Attacks} + \text{Goal Attempts}) / (\text{Tackles} + \text{Interceptions} + \text{Shots Blocked})$$



- Expected Goals: A weighted measure of scoring opportunities. Different from the world-wide used xG metric.  
Expected Goal - home =  $(0.3 * \text{Shots on Target - home}) + (0.15 * \text{Shots Insidebox - home}) + (0.05 * \text{Shots Outsidebox - home})$
- Attack Momentum: Measures the offensive pressure over the last 10 minutes. Indicates the closeness to the score.  
Attack Momentum =  $\text{Dangerous Attacks (last 10 min)} + (\text{Goal Attempts (last 10 min)} * 2) + (\text{Successful Dribbles (last 10 min)} * 0.5)$
- Defensive Pressure: Quantifies defensive actions relative to the opponent's attacks.  
Defensive Pressure =  $[(\text{Tackles} * 0.2) + (\text{Interceptions} * 0.15) + (\text{Shots Blocked} * 0.1)] / (\text{Opponent's Dangerous Attacks})$
- Set Piece Thread: Weighted contribution of set pieces to goal-scoring opportunities.  
Set Piece Thread =  $(\text{Corners} * 0.15) + (\text{Free Kicks} * 0.3) + (\text{Penalties} * 0.85)$
- Psychological Momentum: Measures the current psychological state based on recent events which affects the players mood.  
Psychological Momentum =  $(\text{Goals (last 15 min)} * 1) - (\text{Red Cards} * 1) - (\text{Injuries} * 0.5)$
- Weighted Streak: A weighted average of recent match outcomes, more recent have higher weights.

## 2. Directly Selected Features

These features are used to construct the machine learning models directly from data or simple filtering processes.

- Basic Match Statistics: Goals, Shots on Target, Accurate Crosses, etc.
- Event Metrics: Red Cards, Yellow Cards, Penalties, etc.
- Odds Data: 1, X, 2, Raw odds for home win, draw, and away win.
- Also some features such as “**Streak, Weighted Streak**” are chosen from the papers analyzed in literature review. Previous studies that help to develop this analysis are mentioned in the References section.

All features were integrated into the dataset and form a comprehensive feature set. Forward-filling and imputations ensure data consistency while time-ordering preserves the integrity of time-sensitive predictions.

# Methodology

In this section, methodology employed to develop the betting strategy is discussed. The approach is designed with the key considerations of making a single and accurate prediction per match at an optimal time while using appropriate data and avoiding forward-seeing. To use the available data for training the model, preprocessing steps are taken. The given data is split up into test and training data by assigning the matches starting from November 1, 2024, to test, and remaining to training and validation. After that, unusable data points which have matches suspended or stopped are dropped. Missing values within individual matches are filled using forward filling and the remaining initially null values are filled as 0. Probabilities for the outcomes are calculated from the given odds in the dataset. Available length of each match is checked and matches with less than 40 minutes available are dropped for improving the accuracy of the model. Then the team features are created, named as `interested_col` for previous matches for teams having at least 2 or 3 matches until November 1.

After inspecting the literature, we decided to integrate weighted streak and team form features from the reviewed literature to our model. To reflect how recent match results carry greater weight than older ones, we define a Weighted Streak for a team's  $j$ 'th match as follows. We first define `res_p` as the result of the  $p$ 'th match, taking values in  $\{0, 1, 3\}$  for a loss, draw, or win, respectively. Then, for the team's  $j$ 'th match, we consider its previous  $k$  matches, indexed from  $j - k$  to  $j - 1$ . The weight for each match outcome increases linearly from 1 for the oldest match in that window ( $j - k$ ) up to  $k$  for the most recent match ( $j - 1$ ). We sum these weighted outcomes and normalize by the maximum possible weighted sum (i.e., the sum of weights when all outcomes are wins). By weighting recent matches more heavily, the Weighted Streak provides a responsive measure of a team's performance. Small fluctuations in very recent results have a larger impact, reflecting the notion that a recent win or loss affects a team's confidence and tactics more strongly than older outcomes.

Just like the streak feature, form is also a team factor, and it provides great detail about its recent performance. However, unlike the Streak feature, the form feature is able to incorporate a team's performance relative to their opponents. Form feature ensures that a greater coefficient update is provided if a weak team triumphs over a strong team, and vice-versa. In the case of a draw, the Form of a weak team increases while that of a strong team decreases. When one team  $\alpha$  beats another team  $\beta$ , the Form updates can be viewed as the winning team ( $\alpha$ ) stealing a certain fraction ( $\gamma$ ) of the form of the losing team ( $\beta$ ). Since this fraction is directly proportional to the Form magnitude of a team, larger updates are given for beating a stronger team (high Form value) than for beating a weaker team (low Form value). In the case of a draw, a positive update is given to the weaker team and a negative update to the stronger team.

Ratios for the features that are already present in the dataset are calculated and the home team ratios are added. Then the metrics after a 10-minute mark are selected for use since these cumulative metrics will not be meaningful for the first minutes of the game while they are initialized. Null values for the features are checked for the usability of these features. For features having more null values, 3 prior games are used instead of 2 to reduce features with high null values. Ranked probability scores are used to score the

classifications

For the first betting method, the first 10 minutes are considered as initialization of matches and their features, so they are excluded from the betting window. The model then predicts the outcomes of these matches and calculates the expected profit using odds and betting probabilities. A bet with the highest expected profit is selected, if it is positive and in it is the most expected outcome. If no bet fulfills this condition, then no bet is placed. Two versions for betting\_method\_1 are tested. In version 1 of the first betting method, odds and probabilities from matchmaker are dropped and in version 2, they are used. Version 1 without odd information resulted in less overfitting and with less complex parameters and less variance, therefore this version is selected for use.

For the second betting method, just like the first one, the first 10 minutes are considered as initialization and they are excluded from the betting window. The second method uses regression to predict a numeric value representing the goal difference. The betting strategy is then constructed as such: betting on home win if the predicted value is greater than 1, betting on away win if the predicted value is less than -1 and betting on a draw if the predicted value is close to zero. Bets are made only if a clear prediction is made, and profit or loss is calculated based on the match result.

## Results and Analysis

This section summarizes the methodologies employed and the key findings with the insights. The study captures the development and evaluation of the predictive models for soccer match outcomes and a betting strategy using real-time match data.

Two models were utilized, Random Forest and Gradient Boosted Decision Trees, and two primary approaches were analyzed: classification-based outcome prediction and regression-based goal difference prediction.

Hyperparameter optimization via a grid-search-like cross-validation technique is used in the model construction phase. Promising hyperparameters selected for tuning process.

For Random Forest models, candidate parameters for hyperparameter tuning are:

```
rf_cand_params = {  
    'n_estimators': [512],  
    'max_depth': [12, 8, 4],  
    'min_samples_leaf': [32, 16, 8, 4, 2, 1],  
    'max_features': ['sqrt']  
}
```

For Gradient Boosted Decision Trees models, candidate parameters for hyperparameter tuning are:

```
gbdt_cand_params = {  
    'n_estimators': [512],  
    'max_depth': [8, 6],  
    'min_samples_leaf': [128, 64, 16, 4],  
    'max_features': ['sqrt'],  
    'learning_rate': [0.01, 0.005]  
}
```

## Classification-Based Prediction

Objective is to predict the match outcome based on dynamic match statistics and normalized odds.

1. Feature Set:
  - a. The features included both raw and engineered variables.
  - b. Odds are normalized to represent probability values, ensuring compatibility with probabilistic models.
2. Models Used:
  - a. **Random Forest Classifier:** The most influential hyperparameters are tuned such as number of estimators, maximum depth, minimum samples per leaf, and maximum features. Ranked Probability Score (RPS) and accuracy used as an evaluation metric for cross-validation.
  - b. **Gradient Boosted Decision Trees Classification:** In addition to the parameters tuned for RF model, learning rate also tuned for GDBT model.
3. Evaluation Metrics:
  - a. Accuracy: The proportion of correctly classified samples.
  - b. Ranked Probability Score (RPS): Measures the quality of probabilistic predictions across all classes.

4. Results: The table below shows the results of some experiments.

Random Forest Classification with odds:



|    | n_estimators | max_depth | min_samples_leaf | max_features | Test_RPS_Mean | Train_RPS_Mean |
|----|--------------|-----------|------------------|--------------|---------------|----------------|
| 0  | 512          | 12        | 32               | sqrt         | 0.162947      | 0.040509       |
| 1  | 512          | 12        | 16               | sqrt         | 0.164501      | 0.025458       |
| 2  | 512          | 12        | 8                | sqrt         | 0.163678      | 0.017250       |
| 3  | 512          | 12        | 4                | sqrt         | 0.164586      | 0.013506       |
| 4  | 512          | 12        | 2                | sqrt         | 0.164417      | 0.011820       |
| 5  | 512          | 12        | 1                | sqrt         | 0.164392      | 0.011113       |
| 6  | 512          | 8         | 32               | sqrt         | 0.163017      | 0.080791       |
| 7  | 512          | 8         | 16               | sqrt         | 0.162824      | 0.075119       |
| 8  | 512          | 8         | 8                | sqrt         | 0.162904      | 0.072780       |
| 9  | 512          | 8         | 4                | sqrt         | 0.162918      | 0.071398       |
| 10 | 512          | 8         | 2                | sqrt         | 0.163380      | 0.071012       |
| 11 | 512          | 8         | 1                | sqrt         | 0.162594      | 0.070797       |
| 12 | 512          | 4         | 32               | sqrt         | 0.166489      | 0.151663       |
| 13 | 512          | 4         | 16               | sqrt         | 0.166536      | 0.151591       |
| 14 | 512          | 4         | 8                | sqrt         | 0.166528      | 0.151564       |
| 15 | 512          | 4         | 4                | sqrt         | 0.166520      | 0.151549       |
| 16 | 512          | 4         | 2                | sqrt         | 0.166512      | 0.151548       |
| 17 | 512          | 4         | 1                | sqrt         | 0.166511      | 0.151548       |

The accuracy improved slightly to 68% compared to the without odds case. The inclusion of odds as features helped in precision and recall for the home win and away win categories but still struggled with the X (draw) class. While the prediction quality increased, it also showed signs of overfitting to odds, as profitability decreased.

## Random Forest Classification without odds:



|    | n_estimators | max_depth | min_samples_leaf | max_features | Test_RPS_Mean | Train_RPS_Mean |
|----|--------------|-----------|------------------|--------------|---------------|----------------|
| 0  | 512          | 12        | 32               | sqrt         | 0.176089      | 0.040038       |
| 1  | 512          | 12        | 16               | sqrt         | 0.175905      | 0.025404       |
| 2  | 512          | 12        | 8                | sqrt         | 0.175318      | 0.017720       |
| 3  | 512          | 12        | 4                | sqrt         | 0.174674      | 0.013769       |
| 4  | 512          | 12        | 2                | sqrt         | 0.175903      | 0.011850       |
| 5  | 512          | 12        | 1                | sqrt         | 0.175805      | 0.011292       |
| 6  | 512          | 8         | 32               | sqrt         | 0.175774      | 0.082811       |
| 7  | 512          | 8         | 16               | sqrt         | 0.176259      | 0.077381       |
| 8  | 512          | 8         | 8                | sqrt         | 0.175914      | 0.074826       |
| 9  | 512          | 8         | 4                | sqrt         | 0.176040      | 0.073496       |
| 10 | 512          | 8         | 2                | sqrt         | 0.176386      | 0.072928       |
| 11 | 512          | 8         | 1                | sqrt         | 0.175399      | 0.072811       |
| 12 | 512          | 4         | 32               | sqrt         | 0.183398      | 0.163751       |
| 13 | 512          | 4         | 16               | sqrt         | 0.183461      | 0.163704       |
| 14 | 512          | 4         | 8                | sqrt         | 0.183525      | 0.163663       |
| 15 | 512          | 4         | 4                | sqrt         | 0.183506      | 0.163661       |
| 16 | 512          | 4         | 2                | sqrt         | 0.183506      | 0.163659       |
| 17 | 512          | 4         | 1                | sqrt         | 0.183511      | 0.163658       |

The accuracy achieved was 67%, with better precision and recall for class 1 (home win) compared to classes 2 (away win) and X (draw). This suggests that the model predicts home wins more reliably than other outcomes.

## Gradient Boosted Decision Trees Classification:

|    | n_estimators | max_depth | min_samples_leaf | max_features | Test_RPS_Mean | Train_RPS_Mean | learning_rate |
|----|--------------|-----------|------------------|--------------|---------------|----------------|---------------|
| 0  | 512          | 8         | 128              | sqrt         | 0.185817      | 0.006454       | 0.010         |
| 1  | 512          | 8         | 64               | sqrt         | 0.187960      | 0.004077       | 0.010         |
| 2  | 512          | 8         | 16               | sqrt         | 0.186948      | 0.002405       | 0.010         |
| 3  | 512          | 8         | 4                | sqrt         | 0.188278      | 0.001988       | 0.010         |
| 4  | 512          | 6         | 128              | sqrt         | 0.182701      | 0.024577       | 0.010         |
| 5  | 512          | 6         | 64               | sqrt         | 0.181226      | 0.021711       | 0.010         |
| 6  | 512          | 6         | 16               | sqrt         | 0.181212      | 0.019400       | 0.010         |
| 7  | 512          | 6         | 4                | sqrt         | 0.183925      | 0.018890       | 0.010         |
| 8  | 512          | 8         | 128              | sqrt         | 0.178905      | 0.032616       | 0.005         |
| 9  | 512          | 8         | 64               | sqrt         | 0.177774      | 0.025493       | 0.005         |
| 10 | 512          | 8         | 16               | sqrt         | 0.176581      | 0.019431       | 0.005         |
| 11 | 512          | 8         | 4                | sqrt         | 0.176576      | 0.018123       | 0.005         |
| 12 | 512          | 6         | 128              | sqrt         | 0.175856      | 0.064460       | 0.005         |
| 13 | 512          | 6         | 64               | sqrt         | 0.178010      | 0.060101       | 0.005         |
| 14 | 512          | 6         | 16               | sqrt         | 0.177623      | 0.056579       | 0.005         |
| 15 | 512          | 6         | 4                | sqrt         | 0.177769      | 0.055844       | 0.005         |

This model had an overall accuracy of 57%, which is significantly lower than Random Forest models. The GBDT model performed poorly, especially for the X (draw) category, which had very low precision and recall. This suggests that the model struggled to generalize on the test set, even though it showed decent performance during training.

As can be seen, the best case is:

Random Forest classification with without odds information, n\_estimators=512, max\_depth=8, min\_samples\_leaf=16 where the RPS value is 0.1628. The corresponding model is selected as the best RF model.

Even though RPS is a little better compared to RF without odd information, profits are much lower. This indicates that using odds as features to predict match outcomes causes overfitting to odds and not being able to exploit their inefficiencies.

## Regression-Based Prediction

The objective is to predict the goal differences between teams and derive a betting strategy from it.

1. Feature Set: Since the model tries to predict the goal differences between teams, in addition to the features used in the classification case, goal difference was included as the target variable.
2. Models Used:
  - a. **Random Forest Regressor:** Depth, number of estimators, and minimum samples per leaf parameters are tuned.

**b. Gradient Boosted Decision Trees Regressor:** Same parameters in the RF Regression model with an additional, learning rate.

3. Evaluation Metrics: Mean Absolute Error (MAE): Measures the average deviation between predicted and actual goal differences.
4. Results: The following tables show the results of the cross-validations for the approaches.

Random Forest Regressor

|    | n_estimators | max_depth | min_samples_leaf | max_features | Test_MAE | Train_MAE |
|----|--------------|-----------|------------------|--------------|----------|-----------|
| 0  | 512          | 12        | 32               | log2         | 0.761623 | 0.372129  |
| 1  | 512          | 12        | 16               | log2         | 0.761898 | 0.290974  |
| 2  | 512          | 12        | 8                | log2         | 0.762358 | 0.249767  |
| 3  | 512          | 12        | 4                | log2         | 0.763100 | 0.227373  |
| 4  | 512          | 12        | 2                | log2         | 0.762183 | 0.218871  |
| 5  | 512          | 12        | 1                | log2         | 0.760915 | 0.214022  |
| 6  | 512          | 8         | 32               | log2         | 0.774332 | 0.557192  |
| 7  | 512          | 8         | 16               | log2         | 0.776277 | 0.541736  |
| 8  | 512          | 8         | 8                | log2         | 0.779155 | 0.536756  |
| 9  | 512          | 8         | 4                | log2         | 0.773816 | 0.532639  |
| 10 | 512          | 8         | 2                | log2         | 0.774443 | 0.530246  |
| 11 | 512          | 8         | 1                | log2         | 0.772145 | 0.530767  |
| 12 | 512          | 4         | 32               | log2         | 0.849279 | 0.819572  |
| 13 | 512          | 4         | 16               | log2         | 0.848823 | 0.819433  |
| 14 | 512          | 4         | 8                | log2         | 0.848871 | 0.819133  |
| 15 | 512          | 4         | 4                | log2         | 0.848953 | 0.819018  |
| 16 | 512          | 4         | 2                | log2         | 0.848967 | 0.819019  |
| 17 | 512          | 4         | 1                | log2         | 0.848958 | 0.819018  |



## Gradient Boosted Decision Trees Regression



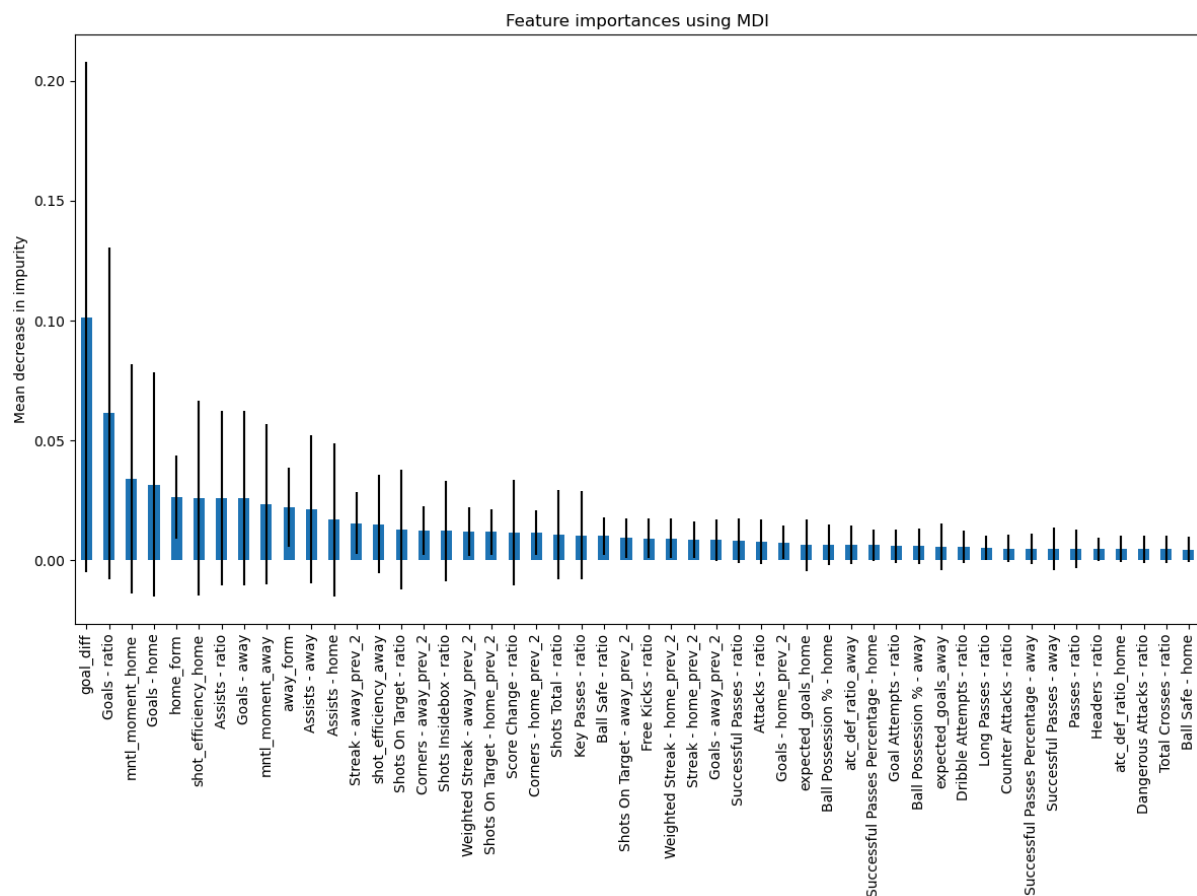
|    | n_estimators | max_depth | min_samples_leaf | max_features | Test_MAE | Train_MAE | learning_rate |
|----|--------------|-----------|------------------|--------------|----------|-----------|---------------|
| 0  | 512          | 8         | 128              | log2         | 0.756996 | 0.294809  | 0.010         |
| 1  | 512          | 8         | 64               | log2         | 0.756730 | 0.244946  | 0.010         |
| 2  | 512          | 8         | 16               | log2         | 0.755901 | 0.208064  | 0.010         |
| 3  | 512          | 8         | 4                | log2         | 0.754531 | 0.199941  | 0.010         |
| 4  | 512          | 6         | 128              | log2         | 0.758128 | 0.433297  | 0.010         |
| 5  | 512          | 6         | 64               | log2         | 0.752091 | 0.411536  | 0.010         |
| 6  | 512          | 6         | 16               | log2         | 0.755396 | 0.399065  | 0.010         |
| 7  | 512          | 6         | 4                | log2         | 0.757080 | 0.396600  | 0.010         |
| 8  | 512          | 8         | 128              | log2         | 0.768470 | 0.450791  | 0.005         |
| 9  | 512          | 8         | 64               | log2         | 0.770183 | 0.407079  | 0.005         |
| 10 | 512          | 8         | 16               | log2         | 0.767209 | 0.375867  | 0.005         |
| 11 | 512          | 8         | 4                | log2         | 0.763500 | 0.368319  | 0.005         |
| 12 | 512          | 6         | 128              | log2         | 0.778236 | 0.572437  | 0.005         |
| 13 | 512          | 6         | 64               | log2         | 0.770165 | 0.557583  | 0.005         |
| 14 | 512          | 6         | 16               | log2         | 0.772083 | 0.549252  | 0.005         |
| 15 | 512          | 6         | 4                | log2         | 0.778014 | 0.547902  | 0.005         |

# Feature Importances

Random forest approaches are performed slightly better than the GBDT models. Even though the performance difference is not showing a high significant difference, using random forest models in classification settings is more advantageous in this specific case.

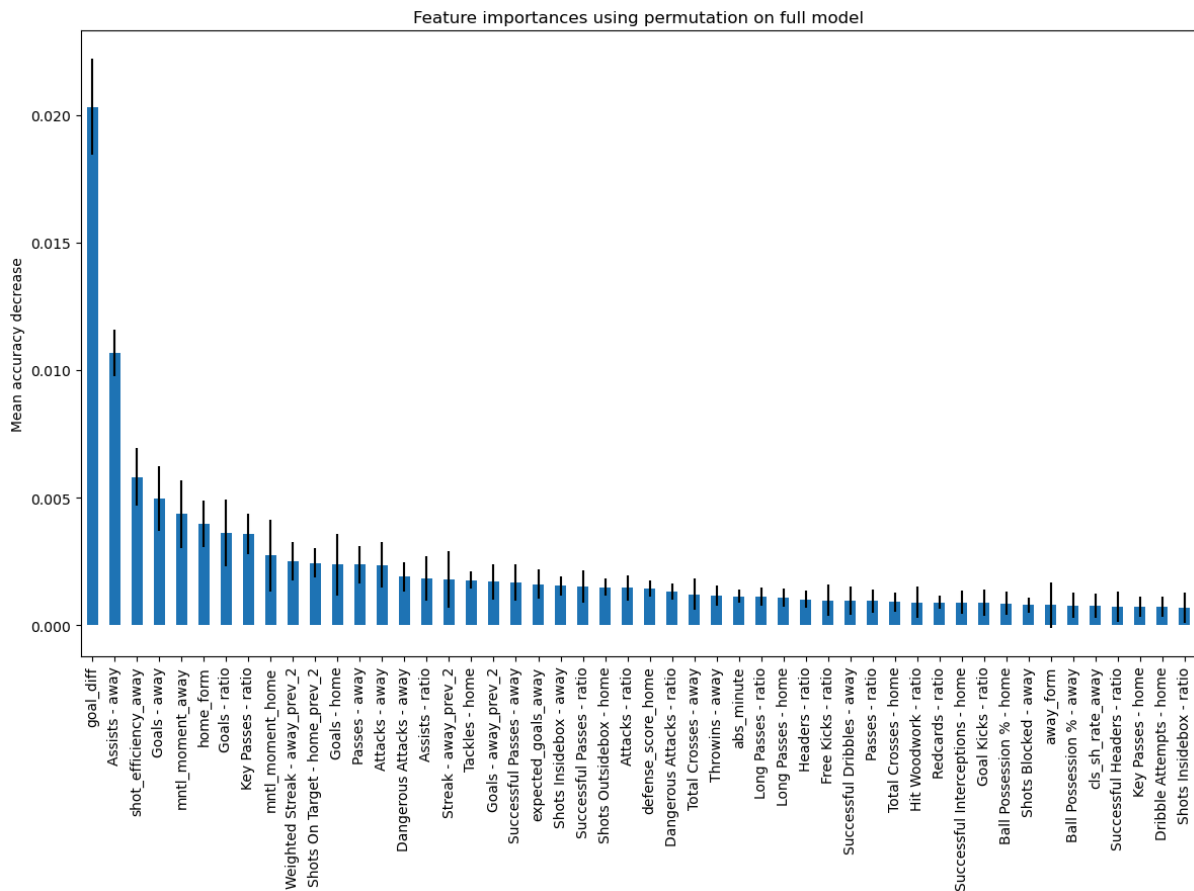
In the literature and industry, there are 2 main methods for assessing feature importances.

1. Mean Decrease in Impurity or Gini Importance, it measures the decrease in Gini impurity in each feature split.



As expected, goal related features, such as goal\_diff, Goals - ratio, Goals - home have significant effects. Besides that, the engineered features, psychological/mental momentum and shot efficiency play an important role to predict the outcomes of the match results. This shows that domain-knowledge and feature extraction is one of the major steps in the study.

2. Permutation Feature Importance, is a model inspection technique that measures the contribution of each feature to a fitted model's statistical performance on a given tabular dataset.



Similar to MDI importances, permutation importances show the role of feature engineering. Extracted features have significance in the permutation importances too. In addition to these extracted features, the “Weighted Streak” feature is chosen from the articles analyzed and mentioned in the literature review section.

Feature importances plots are the evidence of how crucial feature engineering is.

## Betting Strategy Evaluation

Investigating the performed algorithms and implemented techniques, the following table is created as a summary of the findings.

|   | Approach             | Model             | Best Metric | Profit (Units) | Return-to-Bet Ratio |
|---|----------------------|-------------------|-------------|----------------|---------------------|
| 0 | Classification (RPS) | RF (Without Odds) | 0.1747      | 26.81          | 33.94%              |
| 1 | Classification (RPS) | RF (With Odds)    | 0.1629      | 7.82           | 11.67%              |
| 2 | Classification (RPS) | GBDT              | 0.1766      | -10.24         | -9.23%              |
| 3 | Classification (RPS) | Random (Baseline) | 0.2626      | -29.97         | -26.99%             |
| 4 | Regression (MAE)     | RF Regressor      | 0.7609      | 5.70           | 6.78%               |
| 5 | Regression (MAE)     | GBDT Regressor    | 0.7521      | 10.44          | 12.00%              |

- **Random Forest (Without Odds):**

This model achieved the best performance in classification tasks with a profit of 26.81 units and a 33.94% return-to-bet ratio. Its strong profitability demonstrates the model's ability to effectively utilize engineered features, such as psychological momentum and dominance index, without relying on market odds. This allowed the model to identify and exploit inefficiencies in the betting market.

- **Random Forest (With Odds):**

Including odds improved the model's RPS to 0.1549, reflecting better prediction accuracy. However, its profitability dropped to 7.82 units with an 11.67% return-to-bet ratio. The inclusion of odds likely led to over-alignment with market expectations, limiting the model's ability to capitalize on discrepancies between features and odds.

- **GBDT Classification:**

This model achieved a decent RPS (0.2179) but performed really poorly in terms of profitability, resulting in a -10.24 unit loss and a -9.23% return-to-bet ratio. The results highlight the model's sensitivity to feature noise and its inability to balance prediction accuracy with profitable betting strategies.

- **Random Predictions (Baseline):**

Random predictions, used as a baseline, produced a high RPS (0.2626) and suffered significant financial losses (-29.97 units) with a -26.99% return-to-bet ratio. These results underscore the necessity of structured, feature-driven modeling to achieve meaningful and profitable outcomes.

- **Random Forest Regressor:**

In regression tasks, the RF Regressor achieved an MAE of 0.7915 and a profit of 5.70 units, with a 6.78% return-to-bet ratio. Although less profitable than classification models, it provided valuable insights into predicting goal differences and demonstrated reasonable effectiveness in betting applications.

- **GBDT Regressor:**

Among regression models, the GBDT Regressor achieved the best results with an MAE of 0.7945, a profit of 10.44 units, and a 12.00% return-to-bet ratio. Its profitability highlights its ability to capture nuanced relationships between engineered features and match outcomes, making it the strongest regression model in this study.

# Discussion

This study demonstrates implementing machine learning algorithms for predicting soccer games outcomes and designing a betting strategy. By utilizing machine learning models such as Random Forest, Gradient Boosted Decision Trees, and random predictions, the study explores the effectiveness of various approaches in leveraging engineered features, odds data, dynamic match statistics. The discussion focuses on key findings, challenges, and implications of the results.

## 1. Performance of Classification Models

The classification based approach consists of 3 models and the random baseline predictions model. The Random Forest Classification Without Odds data results in the most effective strategy for profitable betting. The profit is 26.81 units and the return to bet ratio is 33.94%. High profit and return ratio indicates that the model captures the underlying market inefficiencies at most. The absence of odds information saves the model from overfitting to bookmaker probabilities. Using Random Forest with classification results in better solutions for every metric compared to Gradient Boosted Decision Trees as expected.

On the other hand, the inclusion of odds information (RF with Odds Data) results in a slight improvement in Ranked Probability Score with the value (0.1549) at the cost of profitability 7.82 units, suggesting that market odds align closely with expected outcomes. While this model is still effective but the generalization ability is less than the previous model and the fact that expected probability is reduced drastically, this model is not the most beneficial option.

Gradient Based Decision Trees Classification model, despite achieving an okay RPS (0.2179), suffered significant losses (-10.24 units). This result underscores the model's sensitivity to feature noise and its inability to balance accuracy and profitability effectively.

The random prediction -baseline- results, with the highest RPS score 0.2626 and the highest loss -29.97 units, highlights that the random prediction is inefficient in this scenario. This specific problem needs special features tailored with the domain knowledge to achieve a sensible result.

## 2. Performance of Regression Models

Regression models were evaluated for predicting goal differences and their application to betting strategies. While classification-based approaches were generally more profitable, the regression approach provided valuable insights into match dynamics and potential score margins.

The Random Forest Regressor achieves a reasonable MAE value 0.7615 and moderate profitability (5.7 units, 6.78% return-to-bet ratio), using random forest regressor is not the best option compared to the results obtained in classification approaches.

The Gradient Boosted Decision Tree Regressor, on the other hand, delivered the best profitability among regression models (10.44 units, 12.00% return-to-bet ratio) and the lowest MAE (0.7945). GDBT regression yields better returns as opposed to the random forest regressor and this is expected for the regression problem.

### 3. Key Challenges and Limitations

**Market Odds and Overfitting:** While including odds improved RPS, it also reduced profitability in some models. This indicates that market odds reflect the collective wisdom of bettors, making it difficult to find significant value.

**Feature Sensitivity:** Engineered features like attack momentum and psychological momentum were critical to model success. However, these features may not generalize well to other datasets or leagues.

**Balancing Accuracy and Profitability:** Models with better accuracy (e.g., lower RPS or MAE) didn't always generate higher profits. This highlights the need to focus on profitability metrics when developing betting strategies.

**Random Baselines:** The poor performance of random baselines across classification and regression tasks underscored the importance of structured modeling. Without data-driven insights, profitability is highly unlikely.

**Data Limitations:** The dataset lacked player-level data, which could have provided more granularity (details) and improved model predictions. Additionally, real-time implementation challenges were not addressed.

### 4. Implications for Real-World Applications

a. **Betting Strategies:** The study compares different machine learning approaches for live betting strategies. Findings state that Models like RF Without Odds can effectively exploit market inefficiencies, offering a robust approach for high-confidence bets.

b. **Feature Engineering:** Dynamic features such as attack momentum and psychological momentum played a significant role in model success. These insights can inform future studies and real-world applications in sports analytics.

c. **Generalizability:** While the study focused on a specific dataset, the methodologies and findings are applicable to other sports or markets with similar dynamics. Future work can evaluate these models across different leagues or seasons, even for different sports.

## 5. Future Work

- Dynamic thresholds can be implemented to explore more live betting strategies. Thresholds shift based on real-time updates to odds and match statistics.
- Integrating detailed player performance metrics could provide additional predictive power to the machine learning algorithms.
- Alternative models can be introduced such as Recurrent Neural Networks (RNNs) or transformers, which may capture internal dependencies more effectively and these consequences may improve predictions.

# Conclusion

The study focuses on constructing machine learning models to predict the results of the soccer matches and designing a live betting strategy. Detailed match statistics data, engineered features, and dynamic odds are used to build models such as Random Forest Regressor/Classifier and Gradient Boosted Decision Trees Regressor/Classifier. Models evaluated for their predictive accuracy, ranked probability scores, and their expected profitability in simulated betting scenarios.

Among the experiments with the constructed models, Random Forest classifier without using odds data achieved the best profit of 26.81 units and a 33.94% return to bet ratio. This model leveraged engineered features like psychological momentum, dominance index, and attack-to-defense ratios to exploit inefficiencies in betting markets, outperforming all other models. Including odds in the feature set improved prediction accuracy (RPS) but reduced profitability, as seen with Random Forest (With Odds), which highlighted the trade-off between alignment with market expectations and the ability to identify high-value bets.

Between 2 regression models, the Gradient Boosted Decision Trees regressor demonstrated the highest profitability 10.44 units and 12% return to bet ratio, with the best MAE 0.7521, shows clearly that the regression models lag behind classification models in the overall profitability.

A consistent challenge across all models was predicting draws (X), which exhibited lower precision and recall compared to home or away wins. This indicates potential gaps in feature representation for such outcomes. Additionally, the poor performance of random baselines underscores the necessity of structured modeling and effective feature engineering.

In conclusion, this study explains the potential of machine learning algorithms to inform profitable betting strategies and provide actionable insights into soccer analytics. The outcomes of the experiments are evidence that the adopted techniques can be generalized to utilize for other sport competitions. By carefully balancing prediction accuracy and market inefficiencies, data-driven approaches like this can serve as a valuable tool for decision-making in sports analytics and betting markets.

# References

1. Baboota, R., & Kaur, H. (2018). Predictive analysis and modelling football results using machine learning approach for English Premier League. *International Journal of Forecasting*, 34(1), 28–43.
2. Berrar, D., Lopes, P., & Dubitzky, W. (2019). Incorporating domain knowledge in machine learning for soccer outcome prediction. *Machine Learning*, 108(1), 97–126.
3. Bunker, R., & Susnjak, T. (2022). The application of machine learning techniques for predicting match results in team sport: A review. *Journal of Artificial Intelligence Research*, 73, 1285–1322.
4. Capobianco, G., Di Giacomo, U., Mercaldo, F., Nardone, V., & Santone, A. (2019). Can machine learning predict soccer match results? In *Proceedings of the 11th International Conference on Agents and Artificial Intelligence (ICAART 2019)* (pp. 458–465). Springer.
5. Horvat, T., & Job, J. (2020). The use of machine learning in sport outcome prediction: A review. *WIREs Data Mining and Knowledge Discovery*, 10(3), e1380.
6. Mandadapu, P. (2023). The evolution of football betting: A machine learning approach to match outcome forecasting and bookmaker odds estimation. *arXiv preprint arXiv:2403.16282*.